FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION

1. Introduction

◈ **Project Title:**

Smart Meet – Video Conferencing Application

2. Project Overview

◈ **Purpose:**

The purpose of Smart Meet is to provide a simple, efficient, and cost-effective platform for online communication, enabling users to conduct video meetings, chat, and collaborate in real time.

◈ **Features:**

- User Registration & Login
- Create & Join Meeting Rooms
- Real-time Video & Audio Calling
- Chat Messaging
- Screen Sharing
- Secure Authentication
- Responsive UI

3. Architecture

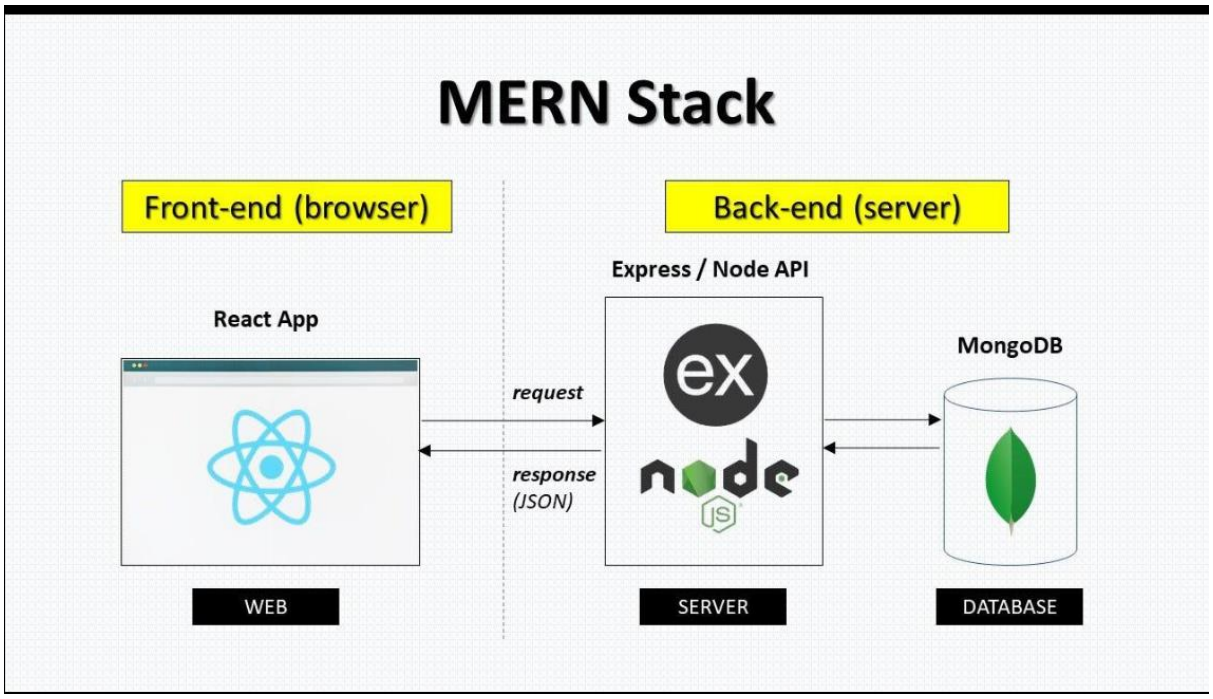
◈ **Frontend (React.js):**

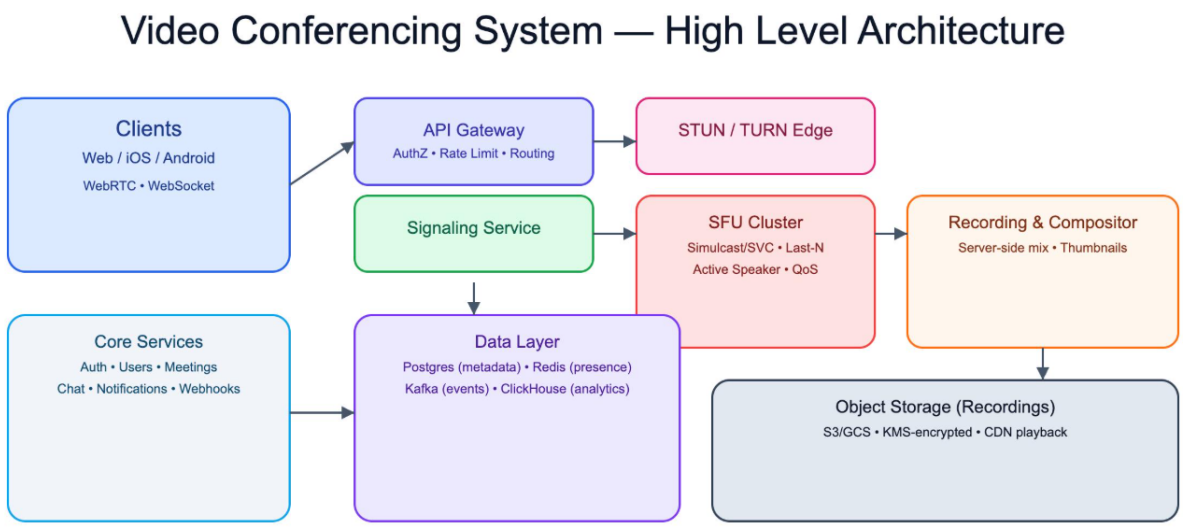
- Built using React.js
- Uses components for UI (Login, Dashboard, Meeting Room)
- Axios for API calls

- Socket.io-client for real-time communication

◈ Backend (Node.js & Express.js):

- REST API using Express.js
- Handles authentication (JWT)
- Manages meeting rooms
- Uses Socket.io for real-time chat





◈ **Database (MongoDB):**

Collections:

- Users (name, email, password)
- Meetings (room ID, participants)

Operations:

- Store user data
- Fetch meeting details
- Authentication validation

4. Setup Instructions

◈ Prerequisites:

- Node.js
- MongoDB
- npm

◈ **Installation Steps:**

1. Clone repository

```
git clone https://github.com/your-repo/smart-meet.git
```

2. Install dependencies

```
cd client
npm install

cd ../server
npm install
```

3. Setup Environment Variables (.env)

MONGO_URI=your_mongodb_url

JWT_SECRET=your_**secret_key**

PORT=5000

5. Folder Structure

◈ Client (Frontend)

client/

```
 ├── src/
 |   ├── components/
 |   ├── pages/
 |   ├── App.js
 |   └── index.js
```

◈ Server (Backend)

server/

```
 ├── routes/
 ├── models/
 ├── controllers/
 ├── server.js
```

6. Running the Application

◈ Frontend:

cd client

npm start

◈ Backend:

cd server

npm start

7. API Documentation

◈ Auth APIs

Method	Endpoint	Description
POST	/api/register	Register user
POST	/api/login	Login user

◈ Meeting APIs

Method	Endpoint	Description
POST	/api/create-room	Create meeting room
POST	/api/join-room	Join meeting

◈ Example Request:

POST /api/login

```
{
  "email": "user@gmail.com",
  "password": "123456"
}
```

◈ Example Response:

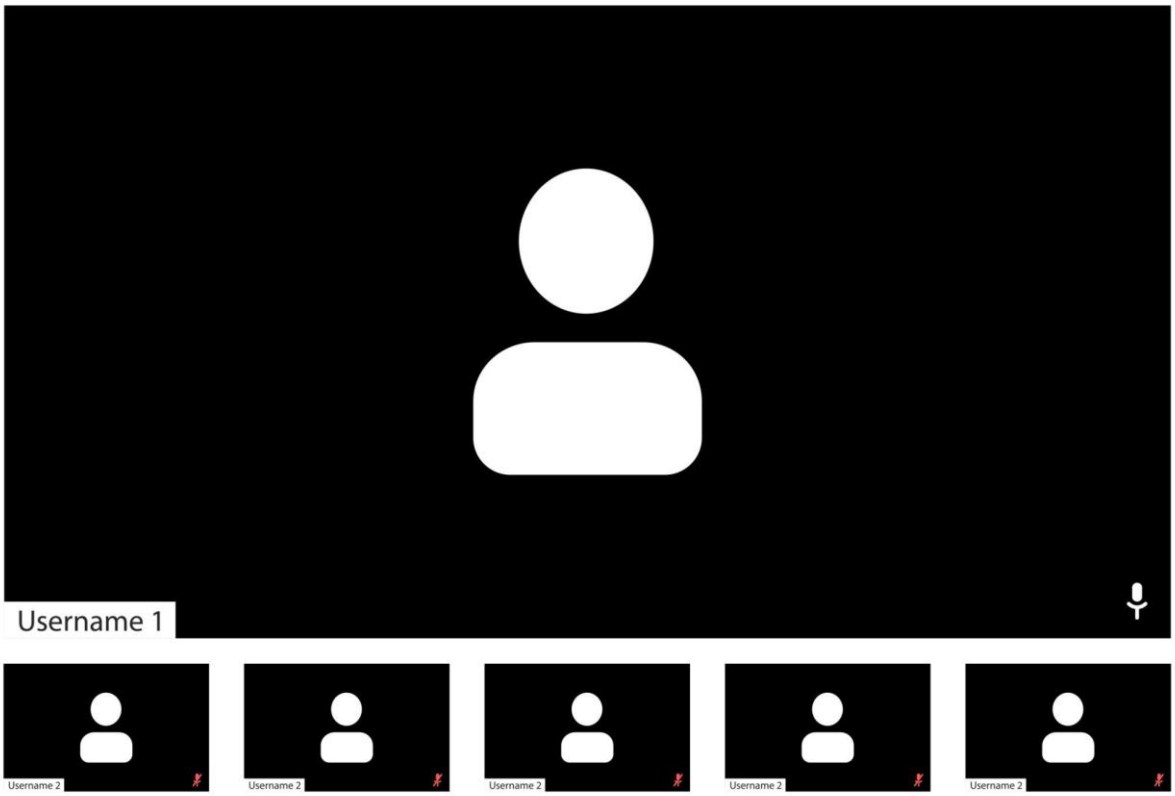
```
{
  "token": "abc123xyz",
  "message": "Login successful"
}
```

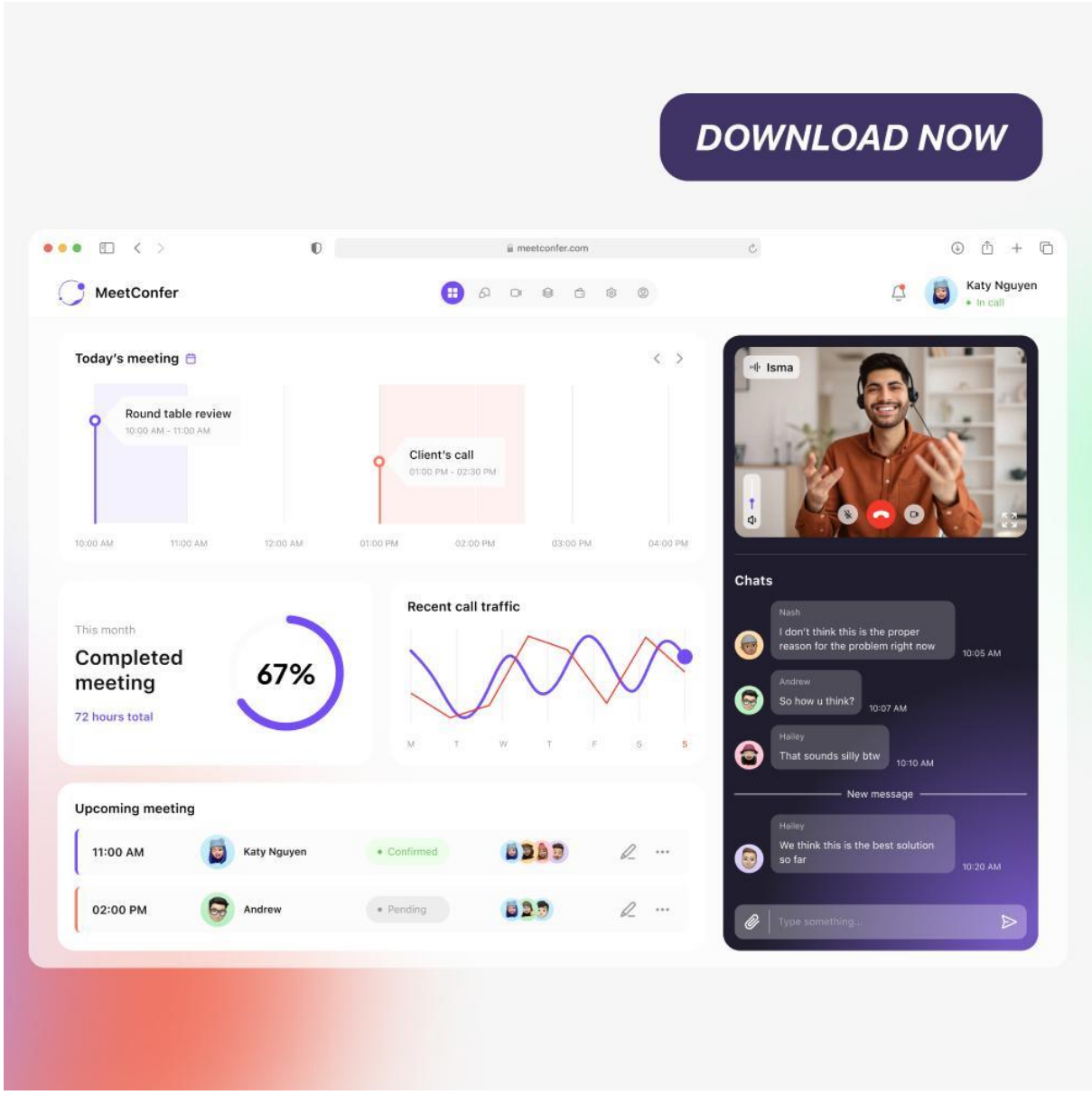
8. Authentication

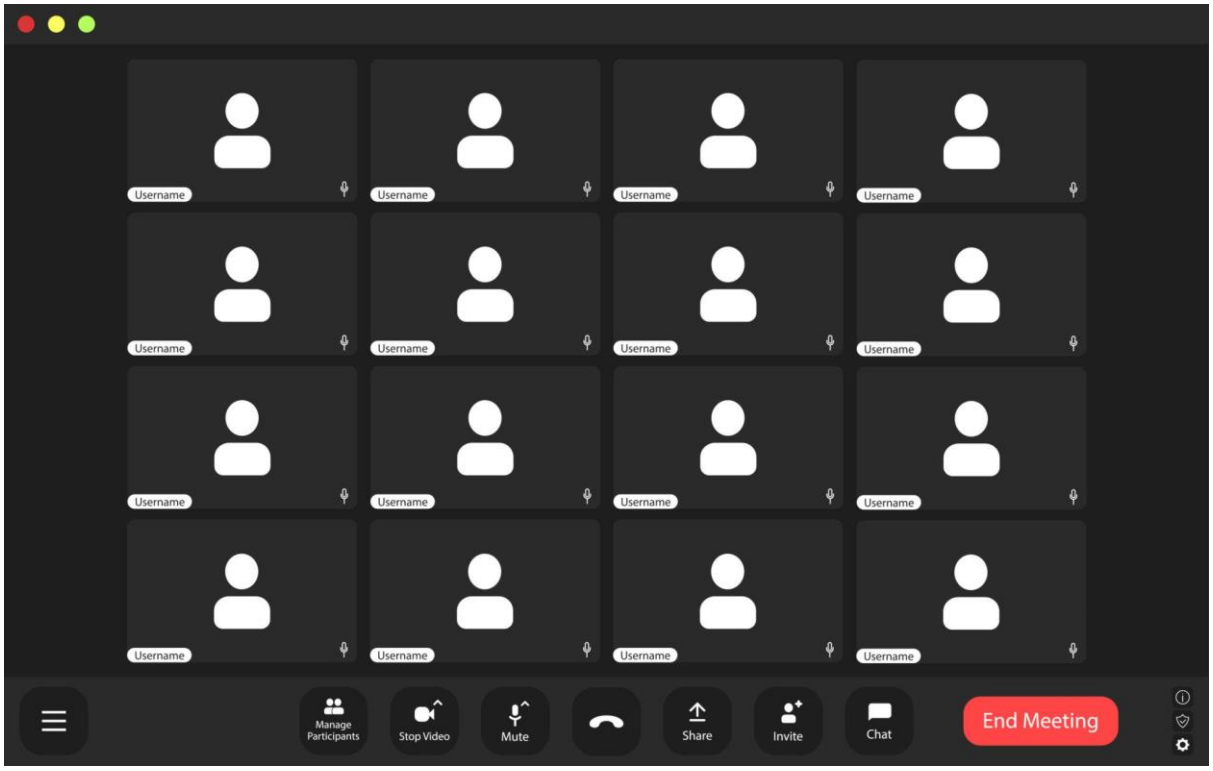
- Uses JWT (JSON Web Token)
- Token generated on login
- Stored in local storage
- Used for secure API access

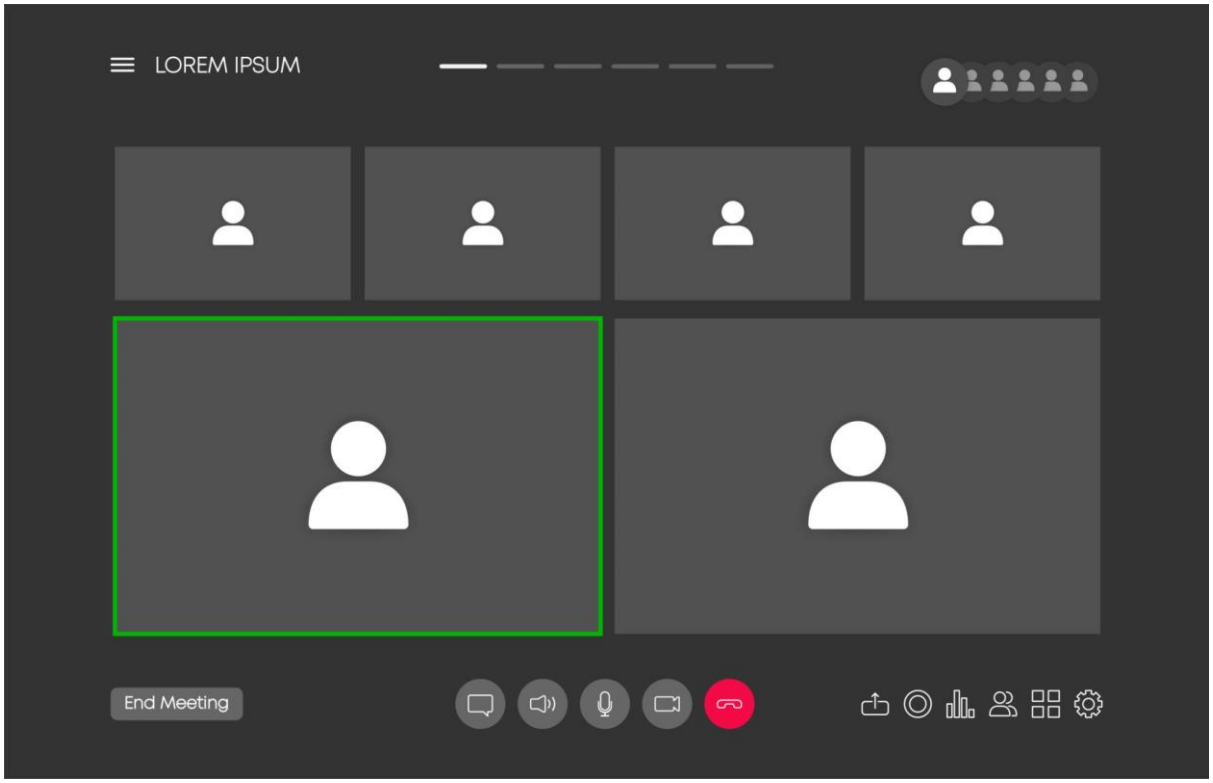
9. User Interface

◈ Key Screens




DOWNLOAD NOW
meetconfer.com
MeetConfer
Katy Nguyen
In call
Today's meeting
Round table review
10:00 AM - 11:00 AM
Client's call
01:00 PM - 02:30 PM
10:00 AM
11:00 AM
12:00 AM
01:00 PM
02:00 PM
03:00 PM
04:00 PM
This month
Completed meeting
72 hours total
67%
Recent call traffic
M
T
W
T
F
S
S
Upcoming meeting
11:00 AM
Katy Nguyen
Confirmed
02:00 PM
Andrew
Pending
Isma
Chats
Nash
I don't think this is the proper reason for the problem right now
10:05 AM
Andrew
So how u think?
10:07 AM
Hailey
That sounds silly btw
10:10 AM
New message
Hailey
We think this is the best solution so far
10:20 AM
Type something...





- **Login Page**
- **Dashboard**
- **Meeting Room**
- **Chat Panel**

10. Testing

◈ Testing Strategy:

- Unit Testing
- Integration Testing
- Manual Testing

◈ Tools Used:

- Postman (API Testing)
- Browser Testing

11. Screenshots / Demo

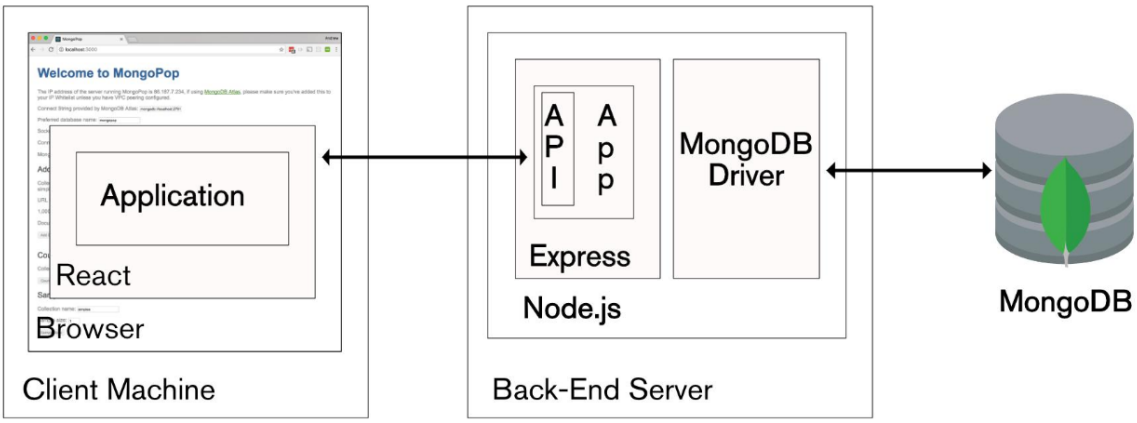
- Screenshots included above
- Demo Link: (Add your deployed link here)

12. Known Issues

- Video lag at high user load
- Minor UI responsiveness issues
- Network dependency for performance

13. Future Enhancements

- Mobile Application (React Native)
- Recording Meetings
- AI Noise Cancellation
- End-to-End Encryption
- Cloud Deployment (AWS)



✅ Conclusion

Smart Meet provides a complete solution for online communication using MERN stack with real-time features and scalable architecture.